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Introduction: Semi-Parametric versus Parametric Estimation in Linear Regression with Two-Piece Normal Errors

Linear regression models traditionally assume Gaussian (symmetric) error distributions and estimate parameters by maximum likelihood. When data exhibit asymmetry or heavy tails, a **two-piece normal (or more generally two-piece location-scale) distribution** provides a flexible parametric alternative by allowing different scales on either side of the mode [Fabrizio Leisen' 2018-02-14].

A fully **parametric** approach treats the shape (e.g., tail) parameter of the two-piece normal as an unknown quantity and estimates it jointly with regression coefficients, often using Bayesian or likelihood-based methods. Recent work has shown that objective Bayesian priors for the tail parameter improve density forecasts, especially in time-series and regression contexts with asymmetric errors [Fabrizio Leisen' 2018-02-14].

In contrast, a **semi-parametric** strategy relaxes the strict distributional assumption on the errors. By leaving the error density unspecified (or modeling it non-parametrically) while retaining a parametric form for the regression component, one can reduce bias arising from misspecified error tails. Semiparametric mixture-regression estimators have been developed that remain identifiable under weaker conditions and possess provable asymptotic properties, even when the true error distribution deviates from normality [Yanyuan Ma' 2018-11-02].

Key Comparisons

Aspect	Parametric (Two-Piece Normal)	Semi-Parametric (Unspecified Errors)
Error specification	Explicit two-piece normal with separate left/right scales; tail parameter estimated	Error density left unspecified; estimated non-parametrically or via mixture
Assumption robustness	Sensitive to misspecification of tail behavior; improves only if two-piece form is appropriate	Robust to a wide range of error shapes, including heavy-tailed or skewed distributions
Estimation method	Maximum likelihood or Bayesian inference with objective priors for tail parameter [Fabrizio Leisen' 2018-02-14]	New semiparametric estimators with established asymptotic normality [Yanyuan Ma' 2018-11-02]
		Holds under weaker conditions;

Identifiability	Requires correct specification of the two-piece family	Holds under weaker conditions; broader identifiability results [Yanyuan Ma' 2018-11-02]
Forecast performance	Demonstrated gains in density forecasts for electricity price data when two-piece model fits well [Fabrizio Leisen' 2018-02-14]	Superior to MLE when normal error assumption is violated; comparable when errors are truly normal [Yanyuan Ma' 2018-11-02]
Computational complexity	Typically lower; closed-form likelihood for two-piece normal	Higher; involves non-parametric components or EM-type algorithms for mixtures

Implications for Practice

- **When asymmetry is mild and the two-piece normal captures the skewness**, a parametric Bayesian approach can yield efficient estimates and better predictive densities, as evidenced in electricity-price applications [Fabrizio Leisen' 2018-02-14].
- **When the error distribution is unknown or potentially far from any two-piece form**, the semi-parametric mixture regression framework offers a safer alternative, delivering consistent estimation without sacrificing efficiency under the true normal case [Yanyuan Ma' 2018-11-02].

Thus, the choice between parametric two-piece normal modeling and semi-parametric error specification hinges on prior knowledge of error asymmetry, the tolerance for model misspecification, and the computational resources available.

Key Topic Groups

Parametric estimation techniques (MLE, Bayesian priors, objective priors) : 1 references found

Parametric estimation techniques—including maximum-likelihood, conventional Bayesian priors, and especially objective (non-informative) priors—provide a rigorous framework for inference in two-piece location-scale models that capture asymmetry in regression errors; recent work has introduced an objective Bayesian methodology for the tail parameter of the skewed exponential power and skewed generalized logistic families, enabling fully parametric treatment of the error distribution in linear regression and time-series contexts and yielding notable gains in density-forecast accuracy, as illustrated with Nordpool electricity price data [Fabrizio Leisen' 2018-02-14].

Notable references:

Loss-based approach to two-piece location-scale distributions with applications to dependent data (14 лют. 2018 р.) ›

Semi-parametric estimation frameworks (mixture models, non-parametric density estimation) : 1 references found

Semi-parametric estimation frameworks—particularly mixture models and non-parametric density approaches—offer a flexible alternative to traditional parametric methods by allowing the error distribution to remain unspecified rather than imposing a strict normality assumption. In the context of linear regression mixtures, this flexibility enables the construction of estimators that retain consistency and asymptotic normality while avoiding the bias that arises when the true error distribution deviates from normality; the resulting estimators can be shown to be identifiable under weaker conditions than classical results and to possess provable asymptotic properties that were previously unavailable for many semiparametric mixture regressions [Yanyuan Ma' 2018-11-02]. Simulation studies confirm that these semiparametric estimators outperform maximum-likelihood estimators when the normal error assumption is violated and remain comparable when the errors are indeed normal, illustrating both robustness and efficiency; an empirical analysis of newly collected Equine Infectious Anemia Virus data further demonstrates their practical utility [Yanyuan Ma' 2018-11-02].

Notable references:

Semiparametric Mixture Regression with Unspecified Error Distributions (3 лист. 2018 р.) ›